

```
4. #
5. # Calculate the ratio between them
6. ratio = apple_price / microsoft_price
7. print("The price ratio is:", ratio)
9. # In pairs trading, we track this ratio over time
10. # If it changes a lot, it might be a trading opportunity
```

Mean Reversion Strategies

Mean reversion strategies operate on the principle that asset prices tend to revert to their historical mean over time. When prices deviate significantly from this mean, algorithms can place trades anticipating a return to normal levels.

The dramatic stock market recovery following the March 2020 COVID-19 crash provided mean reversion traders with numerous opportunities. Stocks like Boeing (BA) that fell more than 70% from their highs eventually rebounded significantly, benefiting algorithms designed to identify extreme deviations from historical price patterns.

A very basic example of a mean reversion check:

```
1. # Simple mean reversion check
2. average_price = 100 # Average price over last 30 days
3. current_price = 80 # Current price
4.
5. # Calculate how far we are from the average (percentage)
6. deviation = (current_price - average_price) / average_price * 100
7.
8. # If we're more than 15% below average, consider buying
9. if deviation < -15:
10.     print("Stock is well below average - potential buy signal")
11. # If we're more than 15% above average, consider selling
12. elif deviation > 15:
13.     print("Stock is well above average - potential sell signal")
14. else:
15.     print("Stock is trading near its average")
16.
```